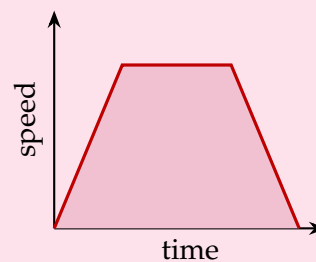
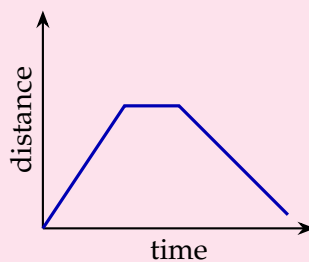
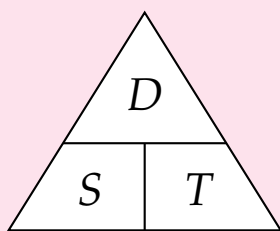

Speed, Distance & Time

Distance–Time & Speed–Time Graphs

Revision Notes, Practice & Homework

Sections 6.3 6.4 7.1 7.2



How to use this worksheet. Read the pink notes first. Each topic has a **class practice** set (white pages, navy headings) and a **homework** set (mint pages with a teal HOMEWORK stripe). Every question page is followed by a **yellow answer page** – only turn over when you have finished. Always show your working and give units. A calculator may be used.

Name: _____ Class: _____ Date: _____

After this practice Actual Exam

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Key notes

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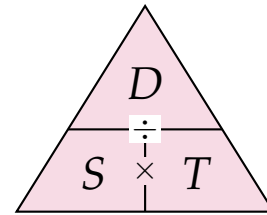
 Class practice (questions)
  Answers (all)
  Homework (questions)

Key Notes

N1 Speed, distance and time

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}} \quad \text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

$$\text{Distance} = \text{Speed} \times \text{Time}$$



- Use the triangle: cover the quantity you want. D on top; S and T underneath (side by side means multiply).
- The units must **match**: km and hours $\rightarrow$ km/h; metres and seconds $\rightarrow$ m/s.
- For a journey at a steady speed, the speed is constant. Otherwise $S = D \div T$ gives the **average** speed.

N2 Working with time

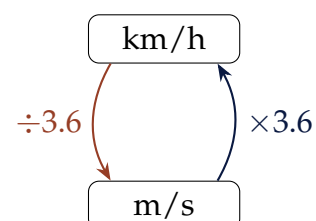
- Minutes $\rightarrow$ hours: **divide by 60**. $24 \text{ min} = \frac{24}{60} = \frac{2}{5} = 0.4 \text{ h}$. Hours $\rightarrow$ minutes: **multiply by 60**. $0.35 \text{ h} = 0.35 \times 60 = 21 \text{ min}$.
- $2 \text{ h } 15 \text{ min} = 2.25 \text{ h}$ (not 2.15 h !). $3.6 \text{ h} = 3 \text{ h} + 0.6 \times 60 \text{ min} = 3 \text{ h } 36 \text{ min}$ (not $3 \text{ h } 60 \text{ min}$).

Minutes	6	10	12	15	20	30	36	40	45	48
Fraction of an hour	$\frac{1}{10}$	$\frac{1}{6}$	$\frac{1}{5}$	$\frac{1}{4}$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{3}{5}$	$\frac{2}{3}$	$\frac{3}{4}$	$\frac{4}{5}$
Decimal hours	0.1	0.1 $\dot{6}$	0.2	0.25	0.3 $\dot{3}$	0.5	0.6	0.6 $\dot{6}$	0.75	0.8

Tip: when the decimal recurs (20 min, 40 min, 10 min), use the **fraction** instead, e.g. $18 \text{ km in } 20 \text{ min} = 18 \div \frac{1}{3} = 54 \text{ km/h}$.

N3 Units of speed

- Common units: km/h, m/s, km/min, mph.
- $1 \text{ km} = 1000 \text{ m}$ and $1 \text{ h} = 60 \text{ min} = 3600 \text{ s}$.
- **km/h $\rightarrow$ m/s: $\times 1000 \div 3600$, i.e. divide by 3.6.**
 $72 \text{ km/h} = 72 \div 3.6 = 20 \text{ m/s}$.
- **m/s $\rightarrow$ km/h: multiply by 3.6.**
 $15 \text{ m/s} = 15 \times 3.6 = 54 \text{ km/h}$.
- **km/min $\rightarrow$ km/h: $\times 60$.** $0.8 \text{ km/min} = 48 \text{ km/h}$.

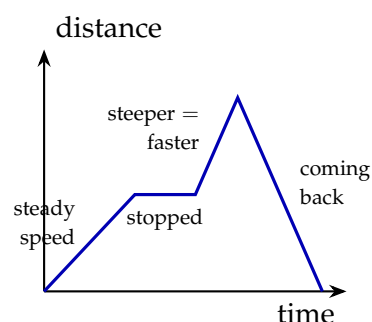


N4 Average speed and harder problems

- **Average speed** = $\frac{\text{total distance}}{\text{total time}}$ (include any time spent stopped). It is **not** the average of the speeds.
e.g. 60 km at 30 km/h (2 h) then 60 km at 60 km/h (1 h): average = $120 \div 3 = 40$ km/h, not 45 km/h.
- **Multi-stage journeys**: work out the distance (or time) of *each* stage first, then add them.
- **Trains and tunnels**: to pass *completely* through a tunnel, the train travels tunnel length + train length. To pass a post or a person, it travels its own length.
- **Required speed**: work out how much time is *left* for the last part, then speed = distance $\div$ time left.
- **Catching up**: two people meet when they have travelled the same distance – compare the times each one took.
- **Miles**: 5 miles = 8 km, so 1 mile = 1.6 km and mph $\times 1.6 =$ km/h. **Time zones**: change both times to the same zone before subtracting.
- **Standard form**: divide the numbers and subtract the powers, e.g. $\frac{6 \times 10^9}{20} = 0.3 \times 10^9 = 3 \times 10^8$ m/s.
- **Algebra**: write each distance as speed $\times$ time, add them, set equal to the total and solve.
e.g. x km/h for 1 h then $(x + 20)$ km/h for 2 h, total 280 km: $x + 2(x + 20) = 280 \Rightarrow 3x = 240 \Rightarrow x = 80$.

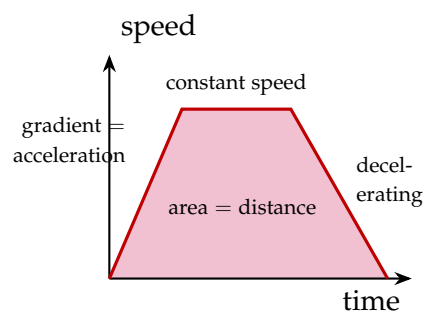
N5 Distance–time graphs

- The **gradient** (steepness) = **speed**
 $= \frac{\text{distance up}}{\text{time across}}.$
- Straight line = constant speed. Steeper = faster.
- **Horizontal** line = **stopped** (not moving).
- Line going **down** = travelling **back** towards the start.
- Read the scale carefully: count how much one small square is worth on each axis.
- Two lines crossing = the two objects are in the **same place at the same time** (they meet / pass).
- Average speed for the whole journey = total distance (there *and* back) $\div$ total time.



N6 Speed–time graphs

- **Acceleration** = **gradient** = $\frac{\text{change in speed}}{\text{time taken}}$.
Units: m/s^2 .
- Line sloping up = accelerating; sloping **down** = **decelerating** (negative acceleration, e.g. -0.75 m/s^2 is a deceleration of 0.75 m/s^2).
- Horizontal line = **constant speed** (acceleration 0) – *not* stopped!
- **Distance travelled** = **area under the graph**.
Split into triangles ($\frac{1}{2}bh$), rectangles and trapezia ($\frac{1}{2}(a+b)h$).
- Check units: if speed is in km/h and time in minutes, change minutes to hours (divide by 60) before finding the area.
- Average speed = total distance (area) $\div$ total time.
- **Curved line**: estimate the area by splitting it into thin strips and treating each as a trapezium.



Common mistakes to avoid

- Writing 1 h 30 min as 1.30 h – it is **1.5 h**. Writing 2.25 h as 2 h 25 min – it is **2 h 15 min**.
- Mixing units: a distance in metres with a time in hours. Convert first.
- Averaging the two speeds instead of using total distance $\div$ total time.
- Forgetting the train's own length when it goes through a tunnel.
- Reading a speed–time graph as if it were a distance–time graph: on a speed–time graph a horizontal line means *steady speed*, on a distance–time graph it means *stopped*.
- Miscalculating the scale – always check what one small square represents.

A Working with time

Sections 6.3

1. [Maximum mark: 6]

Write each time as a fraction of an hour, in its simplest form.

- a) 20 min b) 45 min c) 25 min d) 32 min e) 54 min f) 4 min

2. [Maximum mark: 6]

Change each time into minutes.

- a) 0.35 h b) 0.05 h c) 0.45 h d) $\frac{5}{12}$ h e) $\frac{3}{10}$ h f) 0.7 h

3. [Maximum mark: 4]

Write each time in hours as a decimal (or a mixed number if the decimal recurs).

- a) 4 h 30 min b) 2 h 15 min c) 13 h 15 min d) 1 h 20 min

4. [Maximum mark: 3]

Write each time in hours and minutes. a) 2.75 h b) 3.45 h c) 0.6 h

5. [Maximum mark: 2]

Lizzy left home at 09 30 and arrived at her aunt's house at 11 15. How long did the journey take? Give your answer **a)** in hours and minutes **b)** in hours as a decimal.

6. [Maximum mark: 3]

When the local time in Cairo is 10 00, the local time in Paris is 09 00.

A plane leaves Cairo at 14 20 (Cairo time) and lands in Paris at 17 05 (Paris time).

Work out the flight time.

Answers – A Working with time

worked solutions

1. a) $\frac{20}{60} = \boxed{\frac{1}{3}}$ b) $\boxed{\frac{3}{4}}$ c) $\frac{25}{60} = \boxed{\frac{5}{12}}$ d) $\frac{32}{60} = \boxed{\frac{8}{15}}$ e) $\frac{54}{60} = \boxed{\frac{9}{10}}$ f) $\frac{4}{60} = \boxed{\frac{1}{15}}$
2. a) $0.35 \times 60 = \boxed{21}$ b) $\boxed{3}$ c) $\boxed{27}$ d) $\frac{5}{12} \times 60 = \boxed{25}$ e) $\boxed{18}$ f) $\boxed{42}$ minutes
3. a) $\boxed{4.5}$ h b) $\boxed{2.25}$ h c) $\boxed{13.25}$ h d) $1\frac{20}{60} = \boxed{1\frac{1}{3}}$ h (= 1.33)
4. a) $0.75 \times 60 = 45$: $\boxed{2 \text{ h } 45 \text{ min}}$ b) $0.45 \times 60 = 27$: $\boxed{3 \text{ h } 27 \text{ min}}$ c) $\boxed{36 \text{ min}}$
5. 09 30 to 11 30 is 2 h, so 09 30 to 11 15 is a) $\boxed{1 \text{ h } 45 \text{ min}}$ b) $\boxed{1.75 \text{ h}}$
6. Change the departure to Paris time: 14 20 Cairo = 13 20 Paris.
13 20 to 17 05 = $\boxed{3 \text{ h } 45 \text{ min}}$

Homework A Working with time

Section 6.3

1. [Maximum mark: 5]

Write each time as a fraction of an hour, in its simplest form.

a) 40 min b) 15 min c) 35 min d) 48 min e) 6 min

2. [Maximum mark: 5]

Change into minutes. a) 0.15 h b) 0.8 h c) 0.65 h d) $\frac{7}{12}$ h e) $\frac{1}{20}$ h

3. [Maximum mark: 4]

Write in hours as a decimal. a) 6 h 39 min b) 3 h 24 min c) 15 h 42 min d) 5 h 54 min

4. [Maximum mark: 3]

Write in hours and minutes. a) 1.65 h b) 2.4 h c) 0.85 h

5. [Maximum mark: 2]

Priya leaves at 10 55 and arrives at 12 38. How long is her journey?

6. [Maximum mark: 3]

When the local time in Athens is 09 00, the local time in Berlin is 08 00.

A plane leaves Berlin at 07 40 (Berlin time) and arrives in Athens at 11 15 (Athens time). Find the flight time.

7. [Maximum mark: 2]

A ship leaves port at 20 45. The voyage takes 6 hours 12 minutes. At what time the next day does it arrive?

Homework answers A Working with time

1. a) $\frac{2}{3}$ b) $\frac{1}{4}$ c) $\frac{7}{12}$ d) $\frac{4}{5}$ e) $\frac{1}{10}$
2. a) 9 b) 48 c) 39 d) 35 e) 3 minutes
3. a) $6 + \frac{39}{60} = 6.65$ b) 3.4 c) 15.7 d) 5.9 hours
4. a) $0.65 \times 60 = 39$: 1 h 39 min b) 2 h 24 min c) 51 min
5. 10 55 $\rightarrow$ 11 55 is 1 h; 11 55 $\rightarrow$ 12 38 is 43 min. 1 h 43 min
6. Arrival in Berlin time: 11 15 $-$ 1 h = 10 15. 07 40 to 10 15 = 2 h 35 min
7. 20 45 + 6 h = 02 45; + 12 min = 02 57 the next day.

B Speed, distance and time

Sections 6.3, 6.4

1. [Maximum mark: 10]

Work out the average speed in km/h. Give answers correct to 3 significant figures where needed.

- a) 117 km in 4 h 30 min b) 495 km in 2 h 15 min c) 52 km in 65 minutes
d) 725 km in 1 h 24 min e) 72 km in 1 h 20 min

2. [Maximum mark: 8]

Work out the distance travelled.

- a) 40 km/h for 13 h 15 min b) 852 km/h for 10 h 45 min c) 18 km/h for 55 min d) 650 km/h for 6 h 42 min

3. [Maximum mark: 4]

- a) A plane flies 135 km at an average speed of 180 km/h. How many **minutes** does it take?
b) A train travels 165 km at an average speed of 60 km/h. Give the time in hours and minutes.

4. [Maximum mark: 4]

- a) Liz runs 400 m in 65 seconds. Work out her average speed in m/s (3 s.f.).
b) A car travels 400 m in 18 seconds. Work out its average speed in km/h.

5. [Maximum mark: 3]

Here is part of a train timetable. The 16 05 train from Leeds has an average speed of 120 km/h between Leeds and Newcastle. Work out the distance from Leeds to Newcastle.

Leeds	07 15	09 40	16 05
York	07 42	10 06	16 31
Durham	08 33	10 58	17 24
Newcastle	08 51	11 16	17 41

Answers – B Speed, distance and time

worked solutions

1. a) $117 \div 4.5 = 26 \text{ km/h}$ b) $495 \div 2.25 = 220 \text{ km/h}$ c) $52 \div \frac{65}{60} = 48 \text{ km/h}$

d) $725 \div 1.4 = 517.8 \dots \approx 518 \text{ km/h}$ e) $72 \div 1\frac{1}{3} = 54 \text{ km/h}$

2. a) $40 \times 13.25 = 530 \text{ km}$ b) $852 \times 10.75 = 9159 \text{ km}$ c) $18 \times \frac{55}{60} = 16.5 \text{ km}$ d) $650 \times 6.7 = 4355 \text{ km}$

3. a) $135 \div 180 = 0.75 \text{ h} = 45 \text{ minutes}$ b) $165 \div 60 = 2.75 \text{ h} = 2 \text{ h } 45 \text{ min}$

4. a) $400 \div 65 = 6.153 \dots \approx 6.15 \text{ m/s}$ b) $400 \div 18 = 22.\dot{2} \text{ m/s}; \times 3.6 = 80 \text{ km/h}$

5. $1605 \text{ to } 1741 = 1 \text{ h } 36 \text{ min} = 1.6 \text{ h}$. $120 \times 1.6 = 192 \text{ km}$

B Speed, distance and time – in context

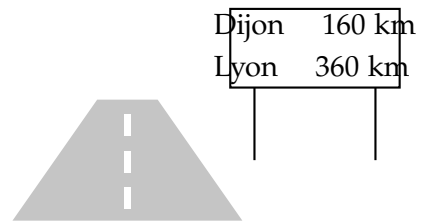
Section 6.4

6.

[Maximum mark: 4]

Emily is driving in France and sees this sign. She drives to Lyon at an average speed of 45 miles per hour. 5 miles = 8 km.

How long should it take her to drive to Lyon?



7.

[Maximum mark: 3]

Maya leaves the gym at 17 25 and drives 18 miles home. The speed limit for the whole journey is 40 mph and she never goes faster than the limit.

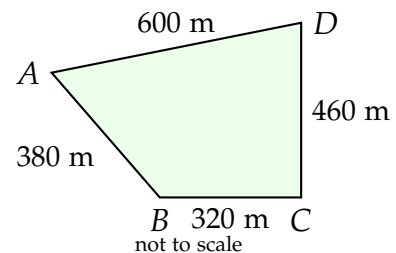
Can Maya get home by 17 50? Give a reason.

8.

[Maximum mark: 4]

Jono runs once around the edge of this field, from A to B to C to D and back to A, at a steady 4 m/s.

How long does it take? Give your answer in minutes and seconds.



9.

[Maximum mark: 4]

A 3-day cycling tour covers 150 km, 120 km and 135 km. The riders' moving times are 6 h 15 min, 5 h 20 min and 5 h 30 min. Work out the average speed for the whole tour.

Answers – B In context

worked solutions

6. $45 \text{ mph} = 45 \times \frac{8}{5} = 72 \text{ km/h}$. Time = $360 \div 72 = \boxed{5 \text{ hours}}$

(Or: $360 \text{ km} = 225 \text{ miles}$, $225 \div 45 = 5 \text{ h}$.)

7. Shortest possible time = $18 \div 40 = 0.45 \text{ h} = 27 \text{ minutes}$. $17 \text{ } 25 + 27 \text{ min} = 17 \text{ } 52$.

No – even at the speed limit she cannot be home before 17 52.

8. Perimeter = $380 + 320 + 460 + 600 = 1760 \text{ m}$. Time = $1760 \div 4 = 440 \text{ s} = \boxed{7 \text{ min } 20 \text{ s}}$

9. Total distance = 405 km . Total time = $6.25 + 5\frac{1}{3} + 5.5 = 17\frac{1}{12} \text{ h}$.

Average speed = $405 \div 17.083\ldots = 23.7\ldots \approx \boxed{23.7 \text{ km/h}}$

Homework B Speed, distance and time

Sections 6.3, 6.4

1. [Maximum mark: 8]

Work out the average speed in km/h, correct to 1 decimal place where needed.

a) 429 km in 6 h 39 min **b)** 433.5 km in 3 h 24 min **c)** 10 260 km in 15 h 42 min

d) A train leaves Prague at 15 20 and arrives in Vienna, 254 km away, at 19 50 the same day.

2. [Maximum mark: 6]

Work out the distance travelled.

a) 80 km/h for 5 h 54 min **b)** 50 km/h, leaving at 07 00 and arriving at 11 15

c) 813 km/h for 11 h 10 min (to the nearest 10 km)

3. [Maximum mark: 3]

Priya drives her van 54 km. She leaves at 10 55 and arrives at 12 38. Calculate her average speed.

4. [Maximum mark: 3]

The 06 07 train from Paris arrives in Montpellier at 09 37. Its average speed is 224 km/h. How far is it from Paris to Montpellier?

5. [Maximum mark: 3]

A train takes 1 h 36 min to travel at an average speed of 85 km/h. A car does the same journey at an average speed of 68 km/h. How much longer does the car take?

Homework answers B Speed, distance and time

1. a) $429 \div 6.65 = 64.51 \dots \approx 64.5 \text{ km/h}$ b) $433.5 \div 3.4 = 127.5 \text{ km/h}$
c) $10\,260 \div 15.7 = 653.50 \dots \approx 653.5 \text{ km/h}$ d) $4 \text{ h } 30 \text{ min} = 4.5 \text{ h}$: $254 \div 4.5 = 56.4 \dots \approx 56.4 \text{ km/h}$
2. a) $80 \times 5.9 = 472 \text{ km}$ b) 4.25 h : $50 \times 4.25 = 212.5 \text{ km}$ c) $813 \times 11\frac{1}{6} = 9081.5 \approx 9080 \text{ km}$
3. $1 \text{ h } 43 \text{ min} = 1\frac{43}{60} \text{ h}$. $54 \div 1.71\dot{6} = 31.45 \dots \approx 31.5 \text{ km/h}$
4. $3 \text{ h } 30 \text{ min} = 3.5 \text{ h}$. $224 \times 3.5 = 784 \text{ km}$
5. Distance = $85 \times 1.6 = 136 \text{ km}$. Car time = $136 \div 68 = 2 \text{ h}$. Extra time = $2 \text{ h} - 1 \text{ h } 36 \text{ min} = 24 \text{ minutes}$

C Converting speeds

Section 6.3

1. [Maximum mark: 4]

Change into m/s. **a)** 144 km/h **b)** 630 km/h **c)** 45 km/h **d)** 105 km/h (2 d.p.)

2. [Maximum mark: 3]

Change into km/h. **a)** 50 m/s **b)** 12.5 m/s **c)** 7.5 m/s

3. [Maximum mark: 3]

- a)** A car travels at 108 km/h for 20 seconds. How many metres does it travel?
b) A car travels at 56 km/h. How long does it take to travel 300 m? Give your answer to the nearest second.

4. [Maximum mark: 3]

An object travels at 2650 metres per second. How many seconds will it take to travel 3.45×10^{10} metres? Give your answer in standard form, correct to 2 significant figures.

5. [Maximum mark: 3]

In May 2019 the distance between Earth and Mars was 3.9×10^7 km.

- a)** A signal travels at 3×10^5 km per second. How long did it take to reach Mars?
b) The signal actually travelled a little slower than this. How does this affect your answer to **a)**?

6. [Maximum mark: 4]

The world speed record for a train is 360 mph. Malcolm drives a train 1 kilometre in 6 seconds. 5 miles = 8 km.

Has Malcolm broken the world record? Show how you decide.

Answers – C Converting speeds

worked solutions

1. a) $144 \div 3.6 = \boxed{40 \text{ m/s}}$ b) $\boxed{175 \text{ m/s}}$ c) $\boxed{12.5 \text{ m/s}}$ d) $105 \div 3.6 = \boxed{29.17 \text{ m/s}}$

2. a) $50 \times 3.6 = \boxed{180 \text{ km/h}}$ b) $\boxed{45 \text{ km/h}}$ c) $\boxed{27 \text{ km/h}}$

3. a) $108 \div 3.6 = 30 \text{ m/s}$. $30 \times 20 = \boxed{600 \text{ m}}$

b) $56 \div 3.6 = 15.5\dot{5} \text{ m/s}$. $300 \div 15.5\dot{5} = 19.28 \dots \approx \boxed{19 \text{ s}}$

4. $3.45 \times 10^{10} \div 2650 = 13\,018\,867.9 \dots \approx \boxed{1.3 \times 10^7 \text{ s}}$

5. a) $\frac{3.9 \times 10^7}{3 \times 10^5} = 1.3 \times 10^2 = \boxed{130 \text{ seconds}}$

b) A slower signal takes **longer**, so the real time was more than 130 s.

6. $1 \text{ km} = \frac{5}{8} \text{ mile} = 0.625 \text{ miles}$. $6 \text{ s} = \frac{6}{3600} = \frac{1}{600} \text{ h}$.

Speed = $0.625 \div \frac{1}{600} = \boxed{375 \text{ mph}}$. $375 > 360$, so **yes**, it is a new record.

C Mixed units & rates

Sections 6.3, 6.4

7. [Maximum mark: 2]

A train travels 24 000 m in 40 minutes. Work out its speed in km/h.

8. [Maximum mark: 2]

A bird flies for 30 minutes at a speed of 7 m/s. How far does it fly, in metres?

9. [Maximum mark: 2]

A car travels at 32 m/s. How many seconds does it take to travel one kilometre?

10. [Maximum mark: 3]

An object travels 49.68 km in 2 hours 18 minutes. Work out its speed in m/s.

11. [Maximum mark: 4]

A machine fills trays of 12 bottles. In one hour it fills a total of 14 400 bottles.
How many seconds does it take to fill one tray?

12. [Maximum mark: 3]

A fuel tank holds 12 000 gallons. Fuel is pumped out at 1500 litres per minute. 1 gallon = 4.5 litres.

How many minutes does it take to empty the tank?

Answers – C Mixed units & rates

worked solutions

7. $24\,000\text{ m} = 24\text{ km}$; $40\text{ min} = \frac{2}{3}\text{ h}$. $24 \div \frac{2}{3} = \boxed{36\text{ km/h}}$

8. $30\text{ min} = 1800\text{ s}$. $7 \times 1800 = \boxed{12\,600\text{ m}}$

9. $1000 \div 32 = \boxed{31.25\text{ s}}$

10. $49.68\text{ km} = 49\,680\text{ m}$; $2\text{ h } 18\text{ min} = 138\text{ min} = 8280\text{ s}$. $49\,680 \div 8280 = \boxed{6\text{ m/s}}$

11. Trays per hour $= 14\,400 \div 12 = 1200$. Time per tray $= 3600 \div 1200 = \boxed{3\text{ s}}$

12. $12\,000 \times 4.5 = 54\,000\text{ litres}$. $54\,000 \div 1500 = \boxed{36\text{ minutes}}$

Homework C Converting speeds

Section 6.3

1. [Maximum mark: 4]

Change into m/s. **a)** 72 km/h **b)** 54 km/h **c)** 900 km/h **d)** 28 km/h (2 d.p.)

2. [Maximum mark: 3]

Change into km/h. **a)** 35 m/s **b)** 4.5 m/s **c)** 220 m/s

3. [Maximum mark: 2]

A car travels at 90 km/h for 45 seconds. How far does it travel, in metres?

4. [Maximum mark: 3]

A rocket travels 100 km at an average speed of 28 440 km/h. How long does this take, in seconds, to the nearest second?

5. [Maximum mark: 4]

a) Light travels 6×10^9 metres in 20 seconds. Calculate the speed of light in m/s, in standard form.**b)** Sound travels 2.06×10^4 metres in 1 minute. Calculate the speed of sound in m/s.

6. [Maximum mark: 3]

A spacecraft travels at 12 km/s. It travels 1.5×10^8 miles. 1 mile = 1.6 km.

How many days does the journey take? Give your answer to the nearest day.

7. [Maximum mark: 2]

A cheetah runs at 28 m/s. A car travels at 95 km/h. Which is faster? Show your working.

Homework answers C Converting speeds

1. a) $\boxed{20}$ b) $\boxed{15}$ c) $\boxed{250}$ d) $28 \div 3.6 = \boxed{7.78} \text{ m/s}$

2. a) $\boxed{126}$ b) $\boxed{16.2}$ c) $\boxed{792} \text{ km/h}$

3. $90 \div 3.6 = 25 \text{ m/s}$. $25 \times 45 = \boxed{1125 \text{ m}}$

4. $100 \div 28\,440 \text{ h} = 0.003516 \dots \text{ h}$; $\times 3600 = 12.66 \dots \approx \boxed{13 \text{ s}}$

5. a) $6 \times 10^9 \div 20 = 0.3 \times 10^9 = \boxed{3 \times 10^8 \text{ m/s}}$ b) $2.06 \times 10^4 \div 60 = 343.3 \dots \approx \boxed{343 \text{ m/s}}$

6. Distance = $1.5 \times 10^8 \times 1.6 = 2.4 \times 10^8 \text{ km}$. Time = $2.4 \times 10^8 \div 12 = 2 \times 10^7 \text{ s}$.
Days = $2 \times 10^7 \div (60 \times 60 \times 24) = 231.4 \dots \approx \boxed{231 \text{ days}}$

7. $28 \times 3.6 = 100.8 \text{ km/h} > 95 \text{ km/h}$, so the **cheetah** is faster.

D Harder speed problems

Section 6.4

1. [Maximum mark: 4]

An aircraft flies at 900 km/h for 3 hours and then at 1100 km/h for 2 hours.

a) Calculate the total distance. b) Calculate the average speed for the whole journey.

2. [Maximum mark: 4]

Sean drives at 60 mph for 2 hours. He then drives the remaining 120 miles at 40 mph.

Sean says, "My average speed was 50 mph." Is he right? Show how you decide.

3. [Maximum mark: 3]

Regan cycles 100 km in 5 hours. His average speed for the first 40 km is 20 km/h.

Work out his average speed for the last 60 km.

4. [Maximum mark: 4]

Sue drives home. She drives 12 km to the motorway in 15 minutes, then 210 km on the motorway at an average speed of 105 km/h, then 8 km from the motorway in 20 minutes.

She stops for a 25-minute rest. Sue leaves at 08 40. What time does she get home?

5. [Maximum mark: 5]

James and Peter cycle the same 40 km route at constant speeds. James takes 2 hours.

Peter starts 6 minutes after James and catches up with him when they have both cycled 12 km.

Work out Peter's speed.

Answers – D Harder speed problems

worked solutions

1. a) $900 \times 3 + 1100 \times 2 = 2700 + 2200 = \boxed{4900 \text{ km}}$ b) $4900 \div 5 = \boxed{980 \text{ km/h}}$
2. Distance = $120 + 120 = 240$ miles. Time = $2 + 120 \div 40 = 2 + 3 = 5$ h.
Average = $240 \div 5 = \boxed{48 \text{ mph}}$, so Sean is **wrong** (he spent longer at the slower speed).
3. First part: $40 \div 20 = 2$ h. Last 60 km takes $5 - 2 = 3$ h. Speed = $60 \div 3 = \boxed{20 \text{ km/h}}$
4. Motorway: $210 \div 105 = 2$ h. Total = $15 + 120 + 20 + 25 = 180$ min = 3 h. 08 40 + 3 h = $\boxed{11 \text{ } 40}$
5. James's speed = $40 \div 2 = 20$ km/h; he takes $12 \div 20 = 0.6$ h = 36 min to cycle 12 km. Peter starts 6 min later, so he takes $36 - 6 = 30$ min = 0.5 h. Speed = $12 \div 0.5 = \boxed{24 \text{ km/h}}$

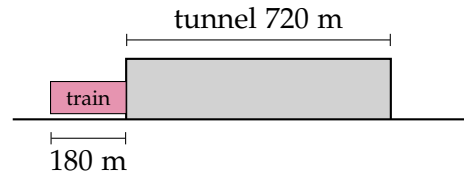
D Trains, bridges and algebra

Section 6.4

6.

[Maximum mark: 3]

A train 180 m long passes completely through a tunnel 720 m long in 36 seconds. Calculate the speed of the train in **a) m/s b) km/h**.



7.

[Maximum mark: 3]

A car of length 4.5 m travels at 90 km/h. It crosses a bridge of length 40.5 m. How long does it take to cross the bridge **completely**?

8.

[Maximum mark: 3]

Zavia drives at x km/h for 2 hours, then at $(x - 15)$ km/h for 3 hours. She travels 405 km altogether. Find x .

9.

[Maximum mark: 6]

One lap of a race track is 6 km. Car *A* takes 2 min 30 s per lap; car *B* takes 2 min 15 s per lap. Both start together from the line *S* and keep constant speeds.

a) Work out the speed of car *A* in km/h.

b) After how long are both cars next at *S* at the same time? **c)** How far has car *A* travelled by then?

Answers – D Trains, bridges and algebra worked solutions

6. Distance = $720 + 180 = 900$ m (the whole train must come out). **a)** $900 \div 36 = \boxed{25 \text{ m/s}}$
b) $25 \times 3.6 = \boxed{90 \text{ km/h}}$
7. Distance = $40.5 + 4.5 = 45$ m. $90 \text{ km/h} = 25 \text{ m/s}$. Time = $45 \div 25 = \boxed{1.8 \text{ s}}$
8. $2x + 3(x - 15) = 405 \Rightarrow 5x - 45 = 405 \Rightarrow 5x = 450 \Rightarrow x = \boxed{90}$
9. **a)** $2 \text{ min } 30 \text{ s} = 150 \text{ s} = \frac{1}{24} \text{ h}$. $6 \div \frac{1}{24} = \boxed{144 \text{ km/h}}$
b) LCM of 150 s and 135 s: $150 = 2 \times 3 \times 5^2$, $135 = 3^3 \times 5$, LCM = $2 \times 3^3 \times 5^2 = 1350 \text{ s}$
= $\boxed{22 \text{ min } 30 \text{ s}}$
c) Car A does $1350 \div 150 = 9$ laps: $9 \times 6 = \boxed{54 \text{ km}}$

Homework D Harder speed problems

Section 6.4

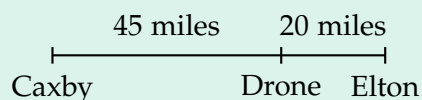
1. [Maximum mark: 4]

A runner runs at 6 m/s for 20 minutes and then at 4 m/s for 40 minutes.

a) Work out the total distance in km. b) Work out the average speed in m/s.

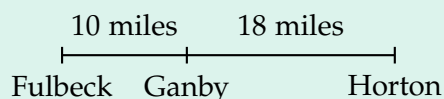
2. [Maximum mark: 3]

Colin drives from Caxby to Drone at an average speed of 30 mph, then from Drone to Elton at 40 mph. Work out his average speed for the whole journey.



3. [Maximum mark: 3]

Raksha leaves Fulbeck at 1000 and drives to Ganby at 40 mph. She then drives on to Horton. She wants to arrive at Horton at 1035. Work out the average speed she must drive from Ganby to Horton.



4. [Maximum mark: 4]

Harry travels from A to B at 40 km/h, then from B to C at 90 km/h. The whole journey takes 4 hours. B to C is 180 km. Work out Harry's average speed for the whole journey.

5. [Maximum mark: 4]

Asif cycles 105 km. He cycles the first 60 km in 2 h 24 min and the last 45 km at 20 km/h. Find his average speed for the whole journey (3 s.f.).

Homework answers D Harder speed problems

1. a) $6 \times 1200 + 4 \times 2400 = 7200 + 9600 = 16\,800 \text{ m} = \boxed{16.8 \text{ km}}$ b) $16\,800 \div 3600 = \boxed{4.67 \text{ m/s}}$
2. Times: $45 \div 30 = 1.5 \text{ h}$, $20 \div 40 = 0.5 \text{ h}$. Average = $65 \div 2 = \boxed{32.5 \text{ mph}}$
3. Fulbeck to Ganby: $10 \div 40 = 0.25 \text{ h} = 15 \text{ min}$, so she reaches Ganby at 10 15.
She has $20 \text{ min} = \frac{1}{3} \text{ h}$ for 18 miles: $18 \div \frac{1}{3} = \boxed{54 \text{ mph}}$
4. $B \rightarrow C$: $180 \div 90 = 2 \text{ h}$, so $A \rightarrow B$ takes 2 h: $40 \times 2 = 80 \text{ km}$.
Average = $(80 + 180) \div 4 = \boxed{65 \text{ km/h}}$
5. Times: 2.4 h and $45 \div 20 = 2.25 \text{ h}$; total 4.65 h. $105 \div 4.65 = 22.58 \dots \approx \boxed{22.6 \text{ km/h}}$

Homework D Trains, bridges and algebra

Section 6.4

6. [Maximum mark: 3]

A train 240 m long passes completely through a tunnel 960 m long in 48 seconds. Find its speed in km/h.

7. [Maximum mark: 4]

A train is 95 m long. It takes 35 seconds to cross a bridge completely at 90 km/h. How long is the bridge?

8. [Maximum mark: 3]

A train 120 m long travels at 144 km/h. It passes under a bridge 20 m wide. How long does it take for the whole train to pass under the bridge?

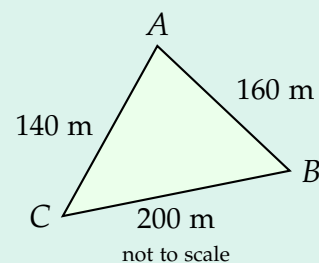
9. [Maximum mark: 4]

A train travels for m minutes at x metres per second.

a) Show that the distance travelled is $\frac{3mx}{50}$ km. b) When $m = 4$ the train travels 9.6 km. Find x .

10. [Maximum mark: 4]

Olav runs around a triangular field. He runs from A to B at 4 m/s, then from B to C at 5 m/s, then from C back to A at a constant speed. His average speed for the whole run is 4 m/s. Work out his speed from C to A .



Homework answers D Trains, bridges and algebra

6. $(960 + 240) \div 48 = 25 \text{ m/s} = 25 \times 3.6 = \boxed{90 \text{ km/h}}$

7. $90 \text{ km/h} = 25 \text{ m/s}$. Distance $= 25 \times 35 = 875 \text{ m} = \text{bridge} + \text{train}$. Bridge $= 875 - 95 = \boxed{780 \text{ m}}$

8. $144 \div 3.6 = 40 \text{ m/s}$. Distance $= 120 + 20 = 140 \text{ m}$. Time $= 140 \div 40 = \boxed{3.5 \text{ s}}$

9. a) Time $= 60m$ seconds, so distance $= 60mx \text{ m} = \frac{60mx}{1000} = \frac{3mx}{50} \text{ km}$. ✓

b) $\frac{3 \times 4 \times x}{50} = 9.6 \Rightarrow 12x = 480 \Rightarrow x = \boxed{40}$

10. Total distance $= 160 + 200 + 140 = 500 \text{ m}$. Total time $= 500 \div 4 = 125 \text{ s}$.

$A \rightarrow B$: $160 \div 4 = 40 \text{ s}$; $B \rightarrow C$: $200 \div 5 = 40 \text{ s}$; so $C \rightarrow A$ takes $125 - 80 = 45 \text{ s}$.

Speed $= 140 \div 45 = 3.11 \dots \approx \boxed{3.11 \text{ m/s}}$

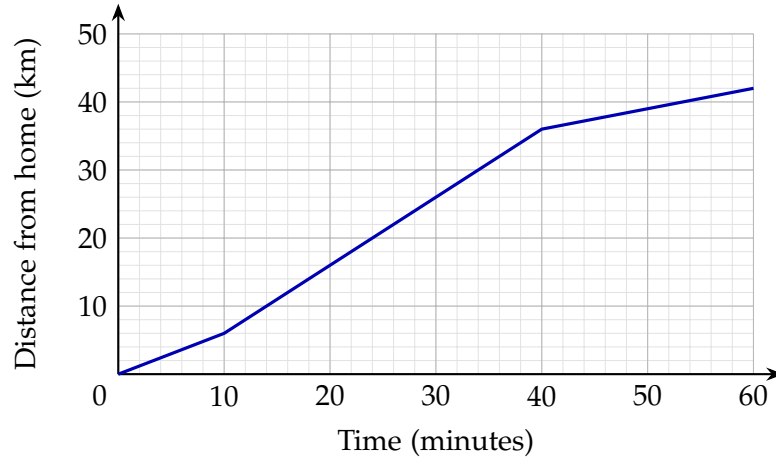
E Distance–time graphs

Section 7.1

1.

[Maximum mark: 7]

The graph shows Paulo's car journey to work.

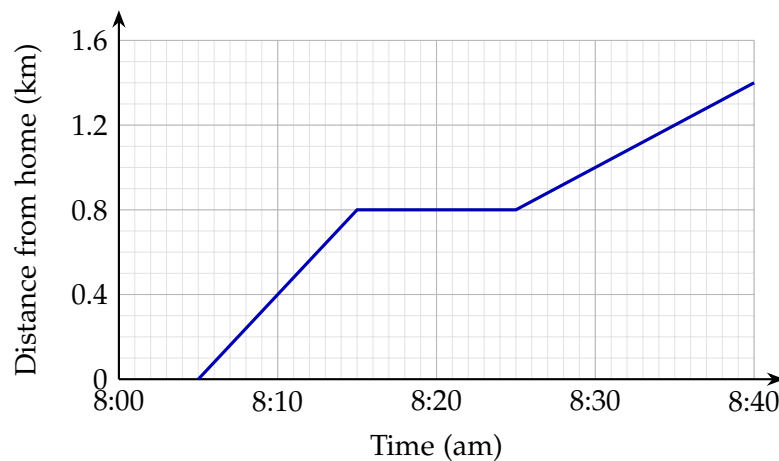


Work out the speed of the car during **a)** the first stage **b)** the second stage **c)** the third stage. Give each answer in **(i)** km/min **(ii)** km/h. **d)** Work out the average speed for the whole journey in km/h.

2.

[Maximum mark: 6]

The graph shows Cara's walk to school. She stops at a friend's house on the way.



- a)** How long does she stop at her friend's house?
b) Work out her speed between 8:05 and 8:15 in km/h. **c)** Work out her speed between 8:25 and 8:40 in km/h.
d) Work out her average speed for the whole walk (from 8:05) in km/h.

Answers – E Distance–time graphs

worked solutions

On the distance axis of Q1 each small square is 2 km; on the time axis each small square is 2 minutes.

- 1.** a) 6 km in 10 min: (i) 0.6 km/min (ii) $0.6 \times 60 = 36 \text{ km/h}$
b) $36 - 6 = 30 \text{ km}$ in 30 min: (i) 1 km/min (ii) 60 km/h
c) $42 - 36 = 6 \text{ km}$ in 20 min: (i) 0.3 km/min (ii) 18 km/h
d) 42 km in 60 min = 42 km/h
- 2.** a) Horizontal from 8:15 to 8:25: 10 minutes
b) 0.8 km in 10 min = 0.08 km/min = 4.8 km/h
c) $1.4 - 0.8 = 0.6 \text{ km}$ in 15 min = 0.04 km/min = 2.4 km/h
d) 1.4 km in 35 min: $1.4 \div \frac{35}{60} = 2.4 \text{ km/h}$

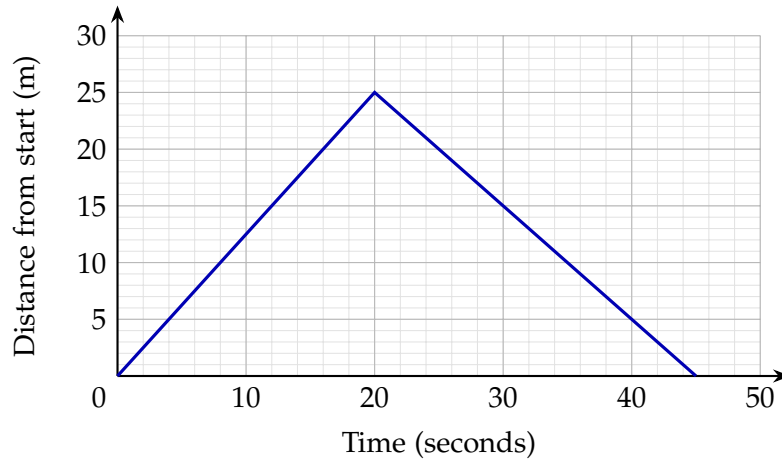
E Distance–time graphs – part 2

Section 7.1

3.

[Maximum mark: 4]

Henri swims a 50 m race in a 25 m pool. The graph shows his distance from the start.

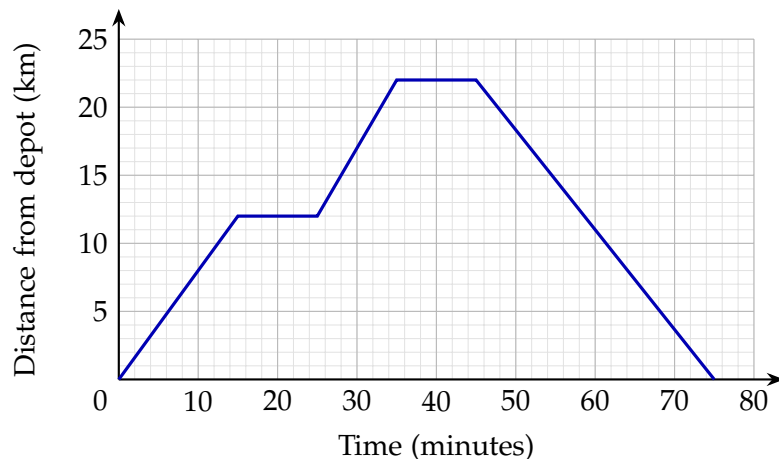


Work out his speed in m/s for **a)** the first length **b)** the second length **c)** the whole 50 m.

4.

[Maximum mark: 6]

A delivery van leaves the depot and returns to it. The graph shows its journey.



- a)** How many times does the van stop, and for how many minutes altogether?
b) On which part of the journey does it travel fastest? **c)** What is the total distance travelled?
d) Work out the average speed for the whole journey in km/h.

Answers – E Distance–time – part 2

worked solutions

3. a) $25 \div 20 = \boxed{1.25 \text{ m/s}}$ b) $25 \div 25 = \boxed{1 \text{ m/s}}$ c) $50 \div 45 = 1.11 \dots \approx \boxed{1.11 \text{ m/s}}$

4. a) $\boxed{2}$ stops (15–25 min and 35–45 min): $\boxed{20 \text{ minutes}}$ altogether.

b) Speeds: 12 km in 15 min = 0.8 km/min; 10 km in 10 min = 1 km/min; 22 km in 30 min = 0.73 km/min.

Fastest is the **second stage (25 to 35 min)** – the steepest line, 60 km/h.

c) 22 km out + 22 km back = $\boxed{44 \text{ km}}$ d) $44 \div \frac{75}{60} = \boxed{35.2 \text{ km/h}}$

E Distance–time graphs – meeting

Section 7.1

5.

[Maximum mark: 8]

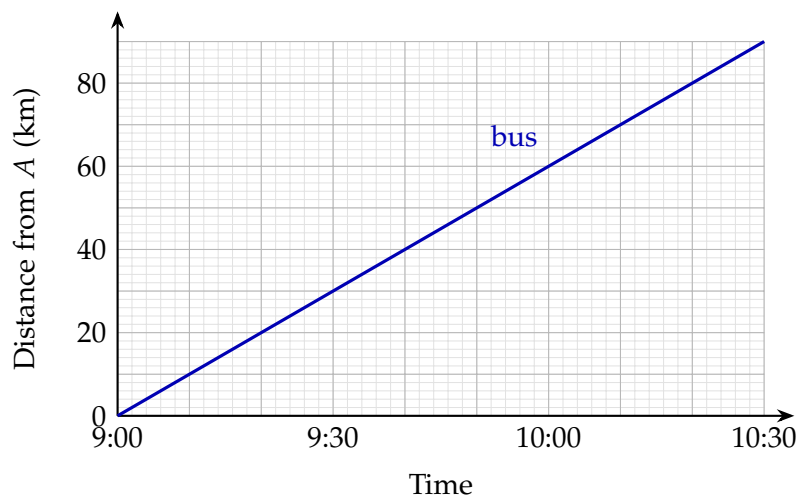
A bus leaves town A at 9:00 and travels at a constant 60 km/h to town B , 90 km away. Its journey is drawn on the graph.

a) Show that the bus arrives at B at 10:30.

At 9:30 a car leaves B and travels towards A at a constant 90 km/h.

b) At what time does the car arrive at A ? **c)** Draw the car's journey on the graph.

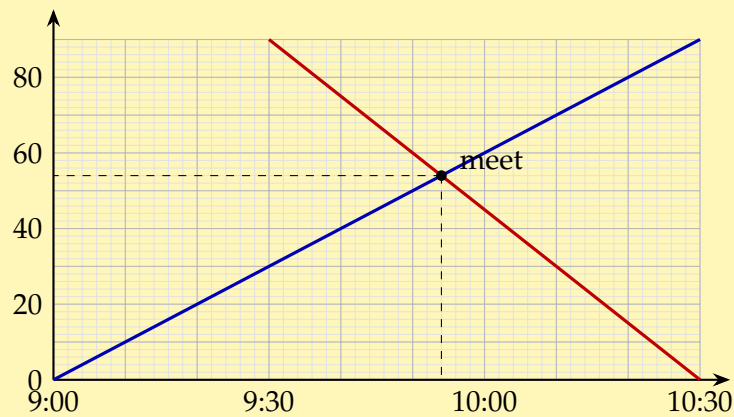
d) Use your graph to find the time when they pass each other, and how far from A this happens.



Answers – E Meeting

worked solutions

5. a) Time = $90 \div 60 = 1.5 \text{ h} = 1 \text{ h } 30 \text{ min}$; $9:00 + 1 \text{ h } 30 \text{ min} = 10:30$. ✓
 b) $90 \div 90 = 1 \text{ h}$: $9:30 + 1 \text{ h} = \boxed{10:30}$
 c) A straight line from (9:30, 90 km) down to (10:30, 0 km) – shown in red.



- d) The lines cross at about $\boxed{9:54}$, $\boxed{54 \text{ km}}$ from A.

Check by algebra: bus = $60t$, car = $90 - 90(t - 0.5)$; $60t = 135 - 90t \Rightarrow t = 0.9 \text{ h} = 54 \text{ min}$.

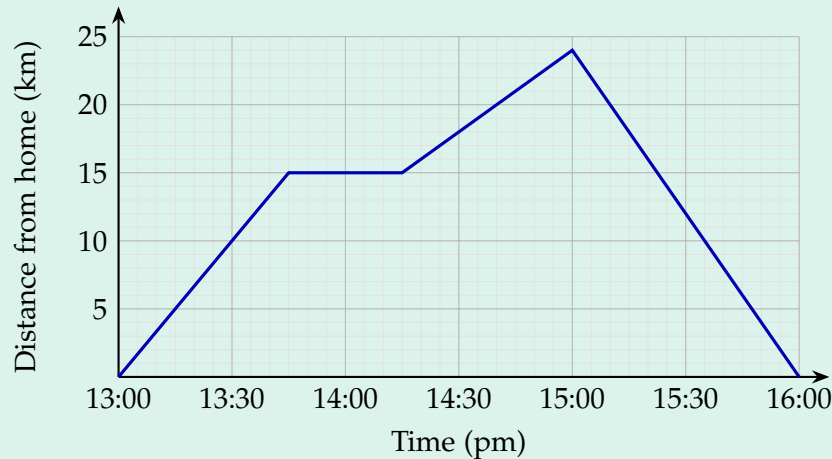
Homework E Distance–time graphs

Section 7.1

1.

[Maximum mark: 7]

Roxy goes on a bike ride and stops once for a rest. The graph shows her journey.

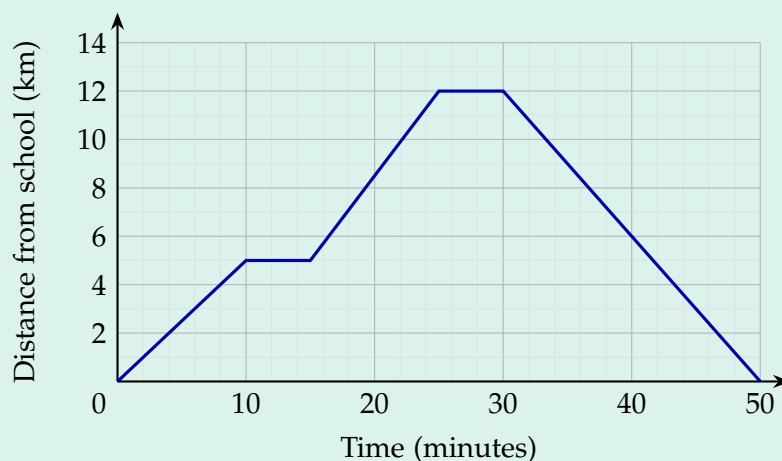


- a) How long does she rest? b) At what time does she get home? c) How far does she ride altogether?
 d) Work out her average speed for the whole ride. e) Work out her fastest speed.

2.

[Maximum mark: 6]

A school bus takes students home. It stops at two villages and then returns to school.



- a) How far are the two villages from the school? b) How long does the bus stop at each village?
 c) Work out the greatest speed of the bus in (i) km/min (ii) km/h.
 d) Work out the average speed for the whole bus journey in km/h.

Homework answers E Distance–time graphs

Q1: each small square is 5 minutes across and 1 km up. Q2: each small square is 2 minutes across and 1 km up.

1. a) 13:45 to 14:15: $\boxed{30 \text{ min}}$ b) $\boxed{16:00}$ c) $24 + 24 = \boxed{48 \text{ km}}$ d) $48 \div 3 = \boxed{16 \text{ km/h}}$

e) Stages: 15 km in 45 min = 20 km/h; 9 km in 45 min = 12 km/h; 24 km in 60 min = 24 km/h.

Fastest = $\boxed{24 \text{ km/h}}$ on the way home (15:00–16:00).

2. a) $\boxed{5 \text{ km}}$ and $\boxed{12 \text{ km}}$ b) $\boxed{5 \text{ minutes}}$ each

c) Speeds: $5 \div 10 = 0.5$; $7 \div 10 = 0.7$; $12 \div 20 = 0.6 \text{ km/min}$. Greatest: (i) $\boxed{0.7 \text{ km/min}}$

(ii) $\boxed{42 \text{ km/h}}$

d) 24 km in 50 min: $24 \div \frac{50}{60} = \boxed{28.8 \text{ km/h}}$

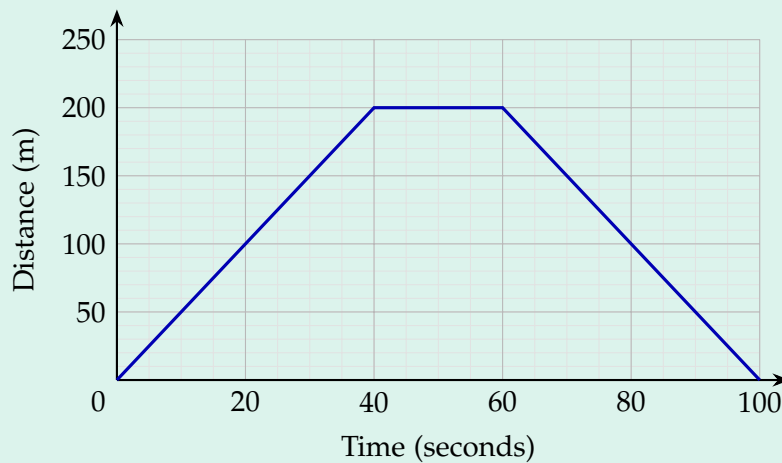
Homework E Reading & drawing graphs

Section 7.1

3.

[Maximum mark: 4]

The graph shows a jogger's distance from a lamp-post.



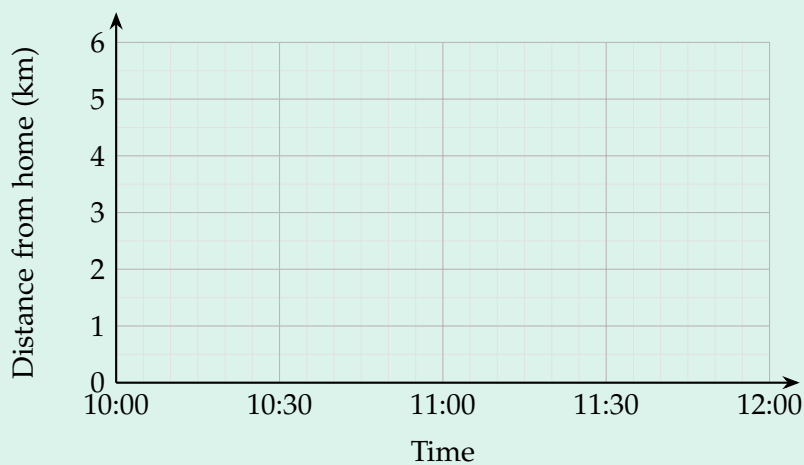
- a) Work out the jogger's speed for the first 40 seconds. b) What happens between 40 and 60 seconds?
 c) Work out the speed on the way back. d) Write the speed in a) in km/h.

4.

[Maximum mark: 4]

Ravi leaves home at 10:00 and walks 3 km at 6 km/h. He rests for 15 minutes, then walks another 2 km in 30 minutes. He then catches a bus straight home at 10 km/h.

- a) Draw Ravi's distance–time graph. b) At what time does he get home?



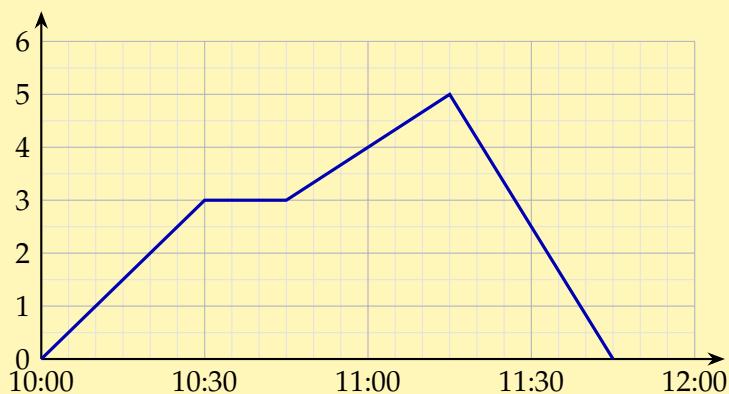
Homework answers E Reading & drawing graphs

3. a) $200 \div 40 = 5 \text{ m/s}$ b) The line is horizontal: the jogger is **stopped** for 20 s.

c) $200 \div 40 = 5 \text{ m/s}$ d) $5 \times 3.6 = 18 \text{ km/h}$

4. Walk 1: $3 \div 6 = 0.5 \text{ h}$, reaching 3 km at 10:30. Rest until 10:45. Walk 2: reaches 5 km at 11:15.

Bus: $5 \div 10 = 0.5 \text{ h}$, so he is home at 11:45.



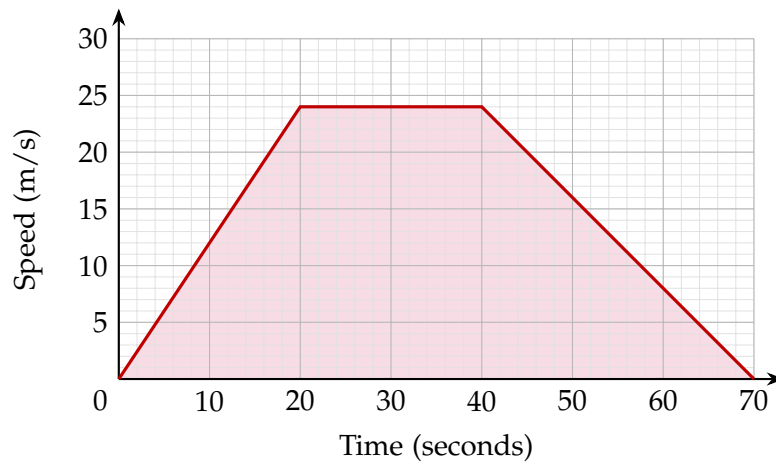
F Speed–time graphs

Section 7.2

1.

[Maximum mark: 5]

The speed–time graph shows a train over 70 seconds.



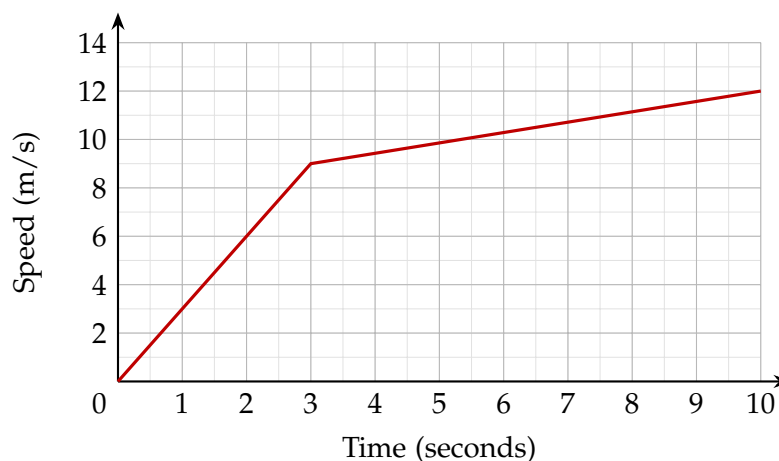
a) Work out the acceleration between 0 and 20 s. **b)** Work out the deceleration between 40 and 70 s.

c) Work out the total distance travelled.

2.

[Maximum mark: 5]

The graph shows a car's speed for the first 10 seconds of a journey.



a) Work out the acceleration between 0 and 3 s, and between 3 and 10 s.

b) Work out the distance travelled in the 10 seconds.

Answers – F Speed–time graphs

worked solutions

1. a) $24 \div 20 = 1.2 \text{ m/s}^2$ b) $24 \div 30 = 0.8 \text{ m/s}^2$

c) Area of trapezium $= \frac{1}{2}(70 + 20) \times 24 = 1080 \text{ m}$

(or triangle 240 + rectangle 480 + triangle 360 = 1080)

2. a) $9 \div 3 = 3 \text{ m/s}^2$; $(12 - 9) \div 7 = 0.428 \dots \approx 0.43 \text{ m/s}^2$

b) Triangle $\frac{1}{2} \times 3 \times 9 = 13.5$; trapezium $\frac{1}{2}(9 + 12) \times 7 = 73.5$. Total = 87 m

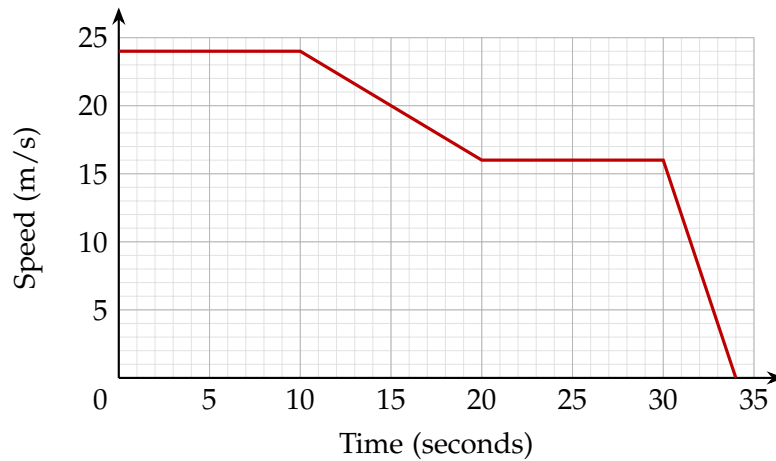
F Speed–time graphs – part 2

Section 7.2

3.

[Maximum mark: 4]

The graph shows the last 34 seconds of a car journey.

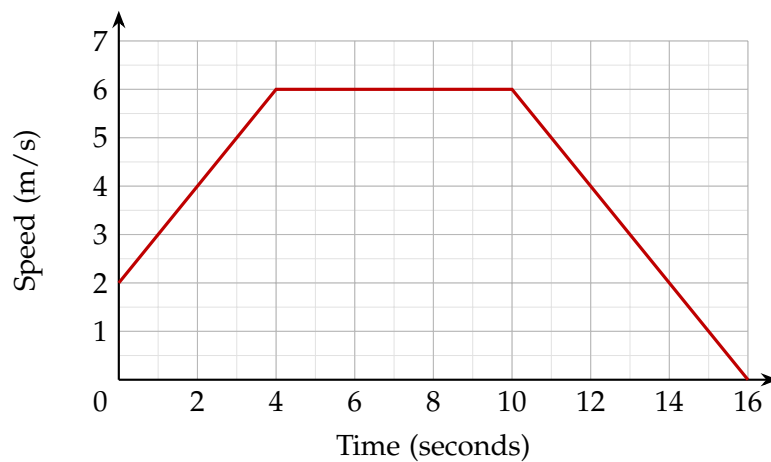


- a)** Find the deceleration of the car as it comes to a stop. **b)** Work out the total distance travelled in the 34 seconds.

4.

[Maximum mark: 4]

The graph shows the last 16 seconds of Roman's bike ride.



- a)** Work out the deceleration. **b)** Work out the total distance Roman travels in the 16 seconds.

Answers – F Speed–time graphs – part 2 worked solutions

3. a) $16 \div 4 = \boxed{4 \text{ m/s}^2}$

b) $24 \times 10 = 240$; $\frac{1}{2}(24 + 16) \times 10 = 200$; $16 \times 10 = 160$; $\frac{1}{2} \times 4 \times 16 = 32$. Total = $\boxed{632 \text{ m}}$

4. a) $6 \div 6 = \boxed{1 \text{ m/s}^2}$

b) Note the graph starts at 2 m/s, not 0! $\frac{1}{2}(2 + 6) \times 4 = 16$; $6 \times 6 = 36$; $\frac{1}{2} \times 6 \times 6 = 18$. Total = $\boxed{70 \text{ m}}$

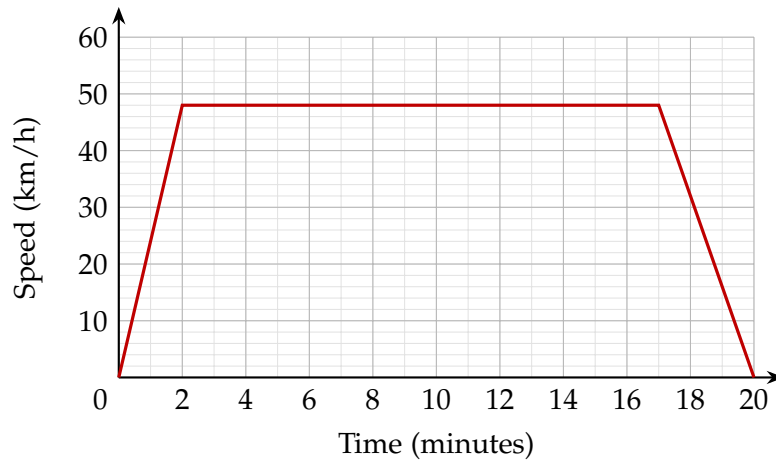
F Mixed units & comparisons

Section 7.2

5.

[Maximum mark: 4]

The graph shows a train journey between two stations. Speed is in km/h and time in minutes.

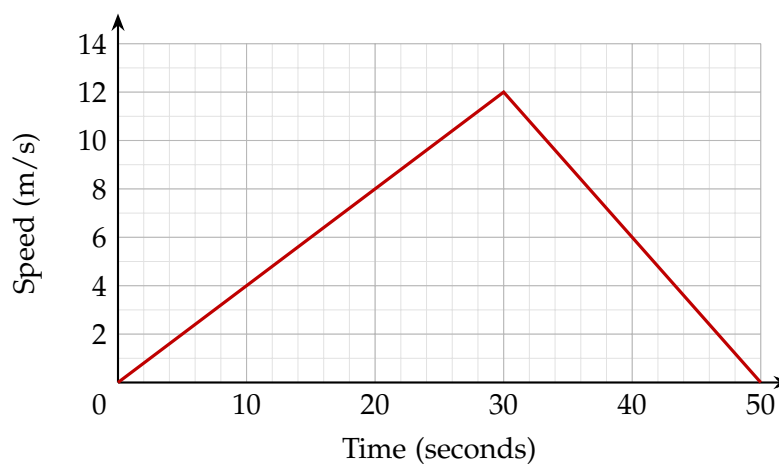


Calculate the distance between the two stations in km.

6.

[Maximum mark: 3]

The graph shows a bus journey between two stops. Hamid runs along the route at a constant 5 m/s. He passes the bus as it leaves the first stop. How far is Hamid from the bus when the bus reaches the second stop?



Answers – F Mixed units & comparisons worked solutions

5. Area = $\frac{1}{2}(20 + 15) \times 48$ where times are in hours: $\frac{1}{2} \times \left(\frac{20}{60} + \frac{15}{60}\right) \times 48 = \frac{1}{2} \times \frac{35}{60} \times 48 =$
14 km

6. Bus distance = $\frac{1}{2} \times 50 \times 12 = 300$ m. Hamid: $5 \times 50 = 250$ m.
Hamid is $300 - 250 =$ 50 m behind the bus.

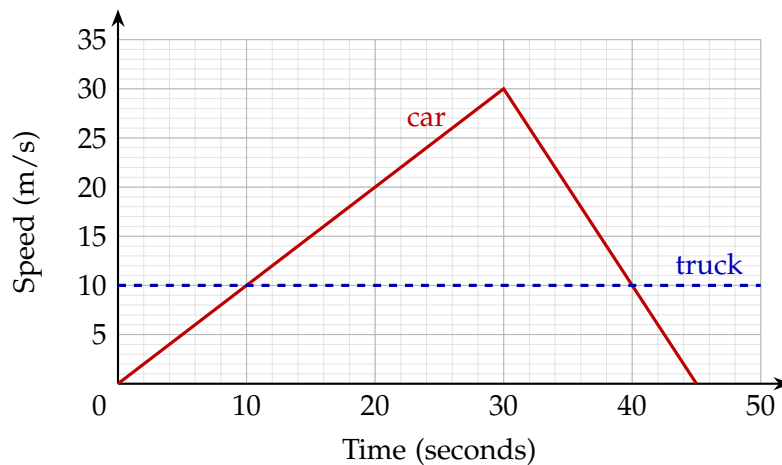
F Two objects & drawing

Section 7.2

7.

[Maximum mark: 5]

The graph shows the speed of a car and of a truck over 45 seconds.



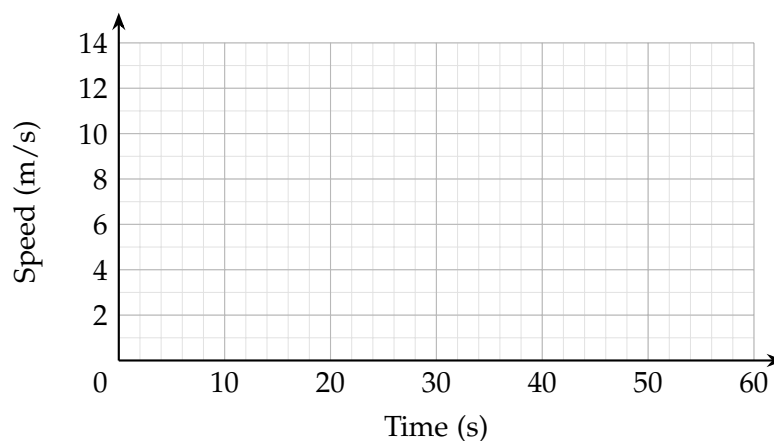
- a) Work out the acceleration of the car over the first 30 s.
 b) Work out the distance the car travels while it is going faster than the truck.

8.

[Maximum mark: 4]

Petra starts from rest and accelerates at a constant 0.5 m/s^2 for 24 seconds. She then travels at a constant speed for 36 seconds.

- a) Draw the speed–time graph for the 60 seconds. b) How far does she travel?



Answers – F Two objects & drawing

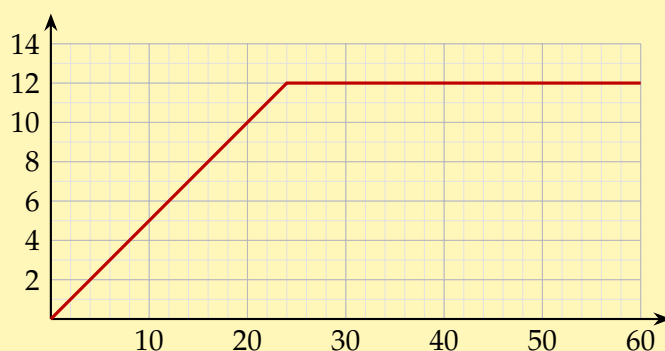
worked solutions

7. a) $30 \div 30 = \boxed{1 \text{ m/s}^2}$

b) Car reaches 10 m/s at $t = 10$ s (accelerating at 1 m/s^2). It decelerates at $30 \div 15 = 2 \text{ m/s}^2$, so it is back to 10 m/s at $t = 30 + 10 = 40$ s.

Area from 10 to 40 s: $\frac{1}{2}(10 + 30) \times 20 + \frac{1}{2}(30 + 10) \times 10 = 400 + 200 = \boxed{600 \text{ m}}$

8. a) Speed after 24 s = $0.5 \times 24 = 12 \text{ m/s}$. Line from (0, 0) to (24, 12), then horizontal to (60, 12).



b) $\frac{1}{2} \times 24 \times 12 + 36 \times 12 = 144 + 432 = \boxed{576 \text{ m}}$

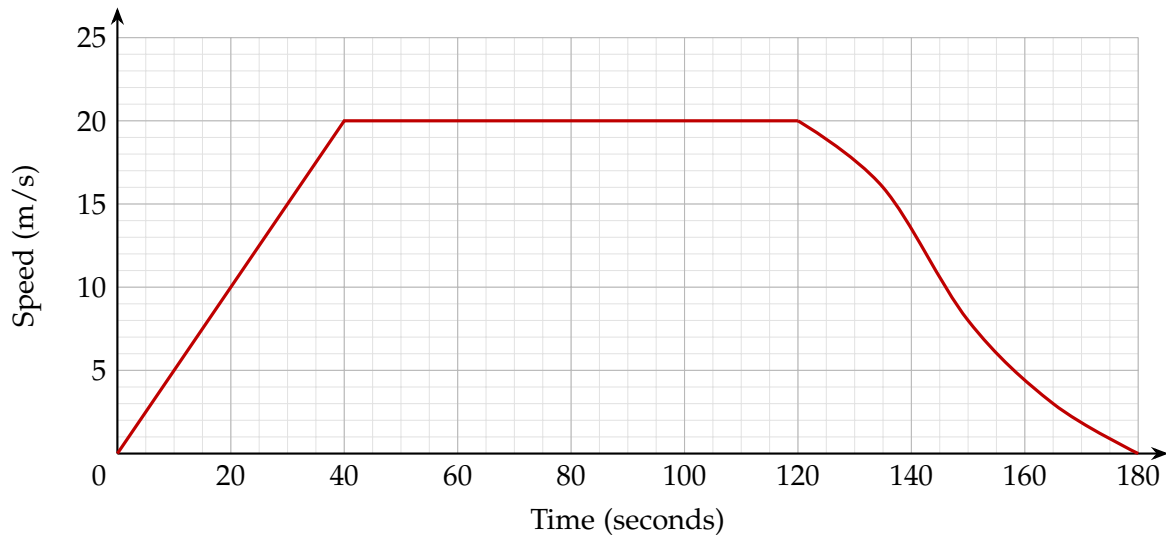
F Estimating area under a curve

Section 7.2

9.

[Maximum mark: 6]

The graph shows the speed of a truck over a 3-minute journey. It travels 2.6 km altogether.



- a) Work out the acceleration between 0 and 40 s. b) Work out the distance travelled in the first 2 minutes.
- c) Estimate the distance travelled while the truck is decelerating. Split the area into four strips 15 s wide.
- d) Use your answer to c) to say whether the 2.6 km total is reasonable.

Answers – F Estimating area under a curve worked solutions

9. a) $20 \div 40 = \boxed{0.5 \text{ m/s}^2}$

b) $\frac{1}{2} \times 40 \times 20 + 80 \times 20 = 400 + 1600 = \boxed{2000 \text{ m}}$

c) Read the speeds at 120, 135, 150, 165, 180 s: about 20, 16, 8, 3, 0 m/s. Use trapezia 15 s wide:

$$15 \left(\frac{20+16}{2} + \frac{16+8}{2} + \frac{8+3}{2} + \frac{3+0}{2} \right) = 15(18 + 12 + 5.5 + 1.5) = \boxed{\approx 555 \text{ m}} \quad (\text{accept about 520–590 m})$$

d) $2000 + 555 \approx 2555 \text{ m} \approx 2.6 \text{ km}$, so **yes**, it is reasonable. (Strips give an estimate, not an exact value – using narrower strips would make it more accurate.)

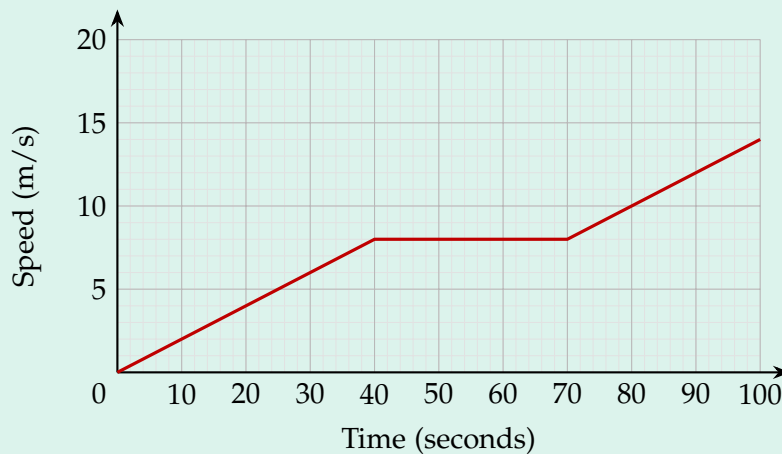
Homework F Speed–time graphs

Section 7.2

1.

[Maximum mark: 6]

The graph shows a cyclist's speed for the first 100 seconds of a journey.



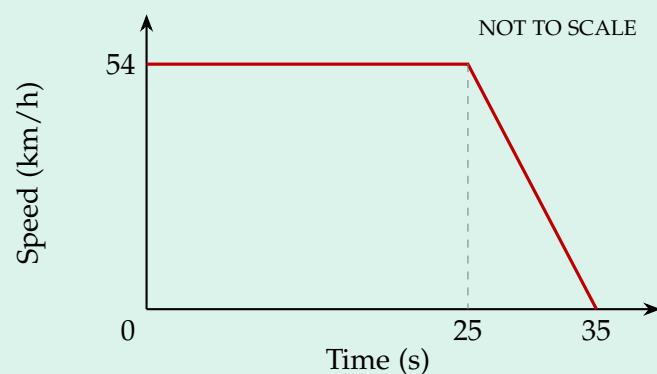
- a) Work out the acceleration between 0 and 40 s, and between 70 and 100 s.
 b) Work out the distance travelled in the first 40 s. c) Work out the total distance in the 100 s.

2.

[Maximum mark: 5]

A car travels at 54 km/h for 25 seconds, then decelerates steadily for 10 seconds until it stops.

- a) Change 54 km/h into m/s.
 b) Find the deceleration in m/s^2 .
 c) Find the distance travelled in the 35 seconds.



Homework answers F Speed–time graphs

1. a) $8 \div 40 = \boxed{0.2 \text{ m/s}^2}$; $(14 - 8) \div 30 = \boxed{0.2 \text{ m/s}^2}$

b) $\frac{1}{2} \times 40 \times 8 = \boxed{160 \text{ m}}$ c) $160 + 8 \times 30 + \frac{1}{2}(8 + 14) \times 30 = 160 + 240 + 330 = \boxed{730 \text{ m}}$

2. a) $54 \div 3.6 = \boxed{15 \text{ m/s}}$ b) $15 \div 10 = \boxed{1.5 \text{ m/s}^2}$ c) $15 \times 25 + \frac{1}{2} \times 10 \times 15 = 375 + 75 = \boxed{450 \text{ m}}$

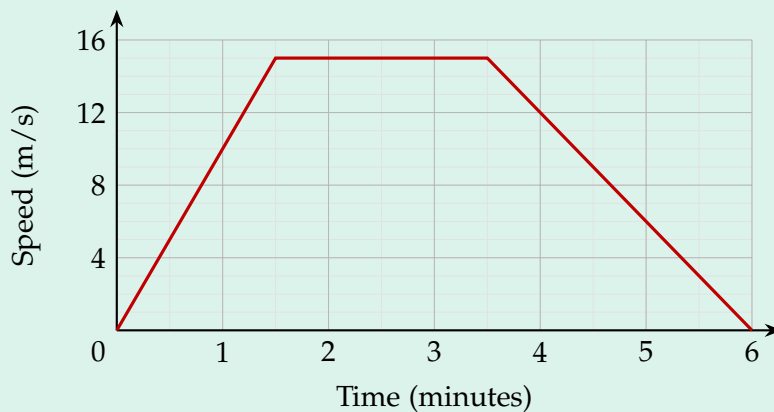
Homework F Speed–time graphs – part 2

Section 7.2

3.

[Maximum mark: 3]

A tram leaves a stop and accelerates for 1.5 minutes until it reaches 15 m/s. It continues at this speed for 2 minutes, then decelerates for 2.5 minutes until it stops.

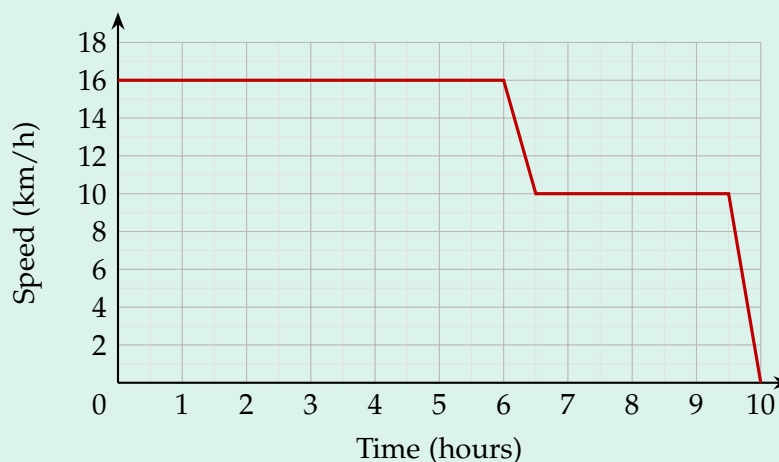


Calculate the distance between the two stops in metres.

4.

[Maximum mark: 5]

A container ship travels at 16 km/h for 6 hours, then slows steadily to 10 km/h over 30 minutes. It travels at 10 km/h for 3 hours, then slows to a stop over 30 minutes.



a) Calculate the total distance travelled. b) Calculate the average speed for the whole voyage.

Homework answers F Speed–time graphs – part 2

3. Change minutes to seconds: 90 s, 120 s, 150 s.

$$\frac{1}{2} \times 90 \times 15 + 120 \times 15 + \frac{1}{2} \times 150 \times 15 = 675 + 1800 + 1125 = \boxed{3600 \text{ m}}$$

4. a) $16 \times 6 = 96$; $\frac{1}{2}(16 + 10) \times 0.5 = 6.5$; $10 \times 3 = 30$; $\frac{1}{2} \times 0.5 \times 10 = 2.5$. Total
 $= \boxed{135 \text{ km}}$

b) $135 \div 10 = \boxed{13.5 \text{ km/h}}$

Homework F Speed–time graphs – part 3

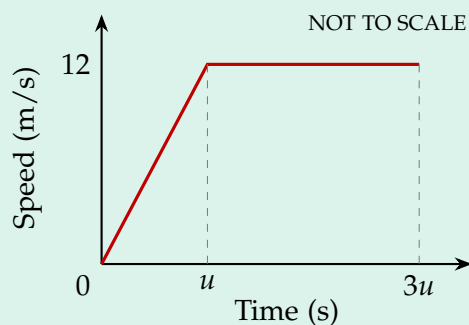
Section 7.2

5.

[Maximum mark: 4]

A car starts from rest and accelerates for u seconds until it reaches 12 m/s. It then travels at 12 m/s for $2u$ seconds. It travels 180 m in the $3u$ seconds.

- a) Find u . b) Find the acceleration in the first u seconds.

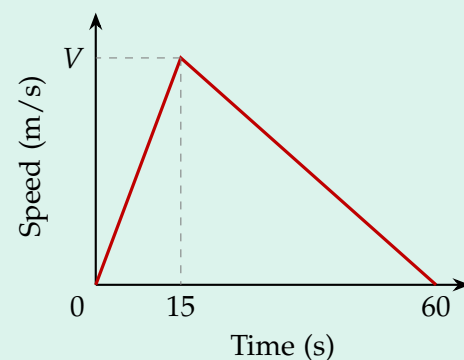


6.

[Maximum mark: 4]

A car accelerates steadily from rest to a top speed of V m/s in 15 s, then decelerates steadily to rest at 60 s. It travels 900 m altogether.

- a) Find V . b) Find the average speed.

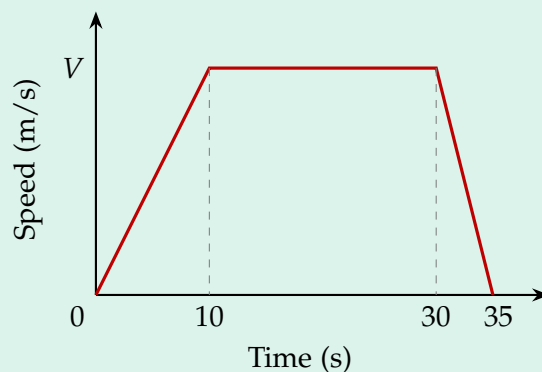


7.

[Maximum mark: 4]

A motorbike accelerates steadily from rest for 10 s, travels at its greatest speed V m/s until 30 s, then decelerates steadily to stop at 35 s. It travels 550 m.

- a) Find V . b) Find the acceleration between 0 and 10 s.



Homework answers F Speed–time graphs – part 3

5. a) $\frac{1}{2} \times u \times 12 + 2u \times 12 = 6u + 24u = 30u = 180 \Rightarrow u = \boxed{6}$ b) $12 \div 6 = \boxed{2 \text{ m/s}^2}$
6. a) $\frac{1}{2} \times 60 \times V = 900 \Rightarrow 30V = 900 \Rightarrow V = \boxed{30}$ b) $900 \div 60 = \boxed{15 \text{ m/s}}$
7. a) Trapezium: $\frac{1}{2}(35 + 20) \times V = 27.5V = 550 \Rightarrow V = \boxed{20}$ b) $20 \div 10 = \boxed{2 \text{ m/s}^2}$

Homework F Drawing & describing

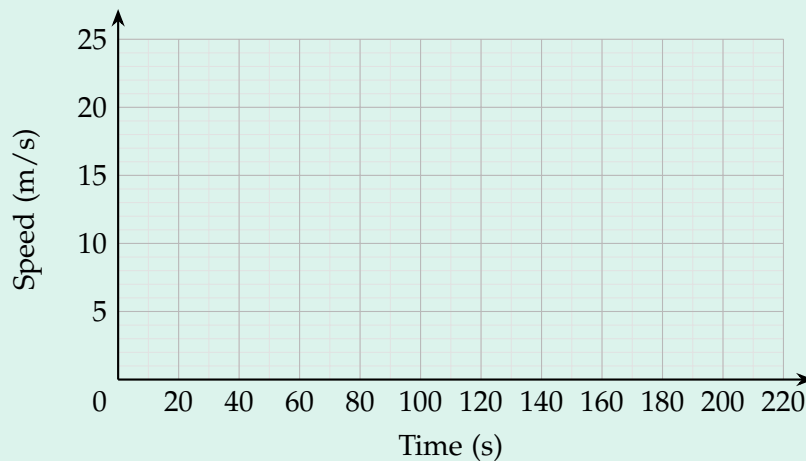
Section 7.2

8.

[Maximum mark: 4]

A train accelerates steadily from rest to 20 m/s in 40 s. It travels at 20 m/s for 2 minutes, then decelerates steadily to rest in 50 s.

a) Draw the speed–time graph. b) Work out the total distance travelled.



9.

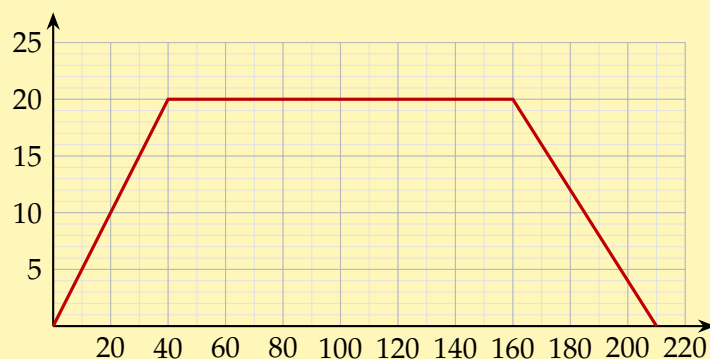
[Maximum mark: 3]

Say what each of these shows. a) A horizontal line on a **distance**–time graph.

b) A horizontal line on a **speed**–time graph. c) A line sloping down on a speed–time graph.

Homework answers F Drawing & describing

8. a) (0, 0) to (40, 20); horizontal to (160, 20); down to (210, 0).



b) $\frac{1}{2} \times 40 \times 20 + 120 \times 20 + \frac{1}{2} \times 50 \times 20 = 400 + 2400 + 500 = \boxed{3300 \text{ m}}$

9. a) The object is **stopped** (stationary). b) Moving at a **constant speed**. c) **Decelerating** (slowing down).

Well done! Go back to any section where you dropped marks, re-read the pink notes for it, and try those questions again.